

Matematikos formulynai

12 klasė

Integralas

- $\int f(x) dx = F(x) + C$, kur $F'(x) = f(x)$
- $\int_a^b f(x) dx = F(b) - F(a)$
- $\int u dv = uv - \int v du$

Nenustatytas integralas

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ ($n \neq -1$)
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int e^x dx = e^x + C$
- $\int \sin x dx = -\cos x + C$, $\int \cos x dx = \sin x + C$
- $\int \frac{1}{\cos^2 x} dx = x + C$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$
- $\int \frac{1}{1+x^2} dx = x + C$

Tikimybė

- $P(A) = \frac{n(A)}{n(\Omega)}$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Nepriklausomi: $P(A \cap B) = P(A) \cdot P(B)$

Kombinatorika

- $n! = 1 \cdot 2 \cdot 3 \cdots n$
- $P_n = n!$
- $A_n^m = \frac{n!}{(n-m)!}$
- $C_n^m = \frac{n!}{m!(n-m)!}$

Statistika

- Vidurkis: $\bar{x} = \frac{1}{n} \sum x_i$
- Dispersija: $D = \frac{1}{n} \sum (x_i - \bar{x})^2$
- Standartinis nuokrypis: $\sigma = \sqrt{D}$